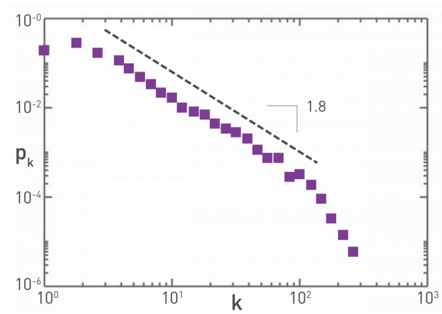
L9: Global Travel Network

Another important factor in the spread of pathogens is the global travel network. Especially with air transportation, in the last few decades, it has become possible for an airborne virus to spread from one point of the planet to all major cities around the world within the first 24 hours.

Imagine, for instance, an infected individual sneezing while waiting at the security control line of a busy airport such as JFK. The passengers around him/her may be traveling to almost every other country on the planet.



The plot at the right (Fig.10.15 from Network Science by Albert-László Barabási) refers to the air transportation network, where the nodes are airports and the links refer to direct flights between airports: the degree distribution of this network is a power-law with an exponent close to 2. Atlanta's airport is one of the most connected in the world and resides at the tail of this distribution.